

PROBABILITY THEORY

MTM2592-Gr:2

SIXTH WEEK

Assc. Prof. Dr. Ulku (BABUSCU) YESIL

ubabuscu@yildiz.edu.tr

ulkubabuscu@gmail.com

<https://avesis.yildiz.edu.tr/ubabuscu>

MULTIVARIATE DISTRIBUTIONS

In this section, we will first deal with the bivariate case. Then, we will extend the analysis to the multivariate case.

EXAMPLE :

Two caplets are selected at random from a bottle containing 3 aspirin, 2 sedative, and 4 laxative caplets. If X and Y are, respectively, the numbers of aspirin and sedative caplets included among the 2 caplets drawn from the bottle, find the probabilities associated with all possible pairs of values of X and Y .

JOINT PROBABILITY DISTRIBUTION:

DEFINITION **JOINT PROBABILITY DISTRIBUTION.** *If X and Y are discrete random variables, the function given by $f(x, y) = P(X = x, Y = y)$ for each pair of values (x, y) within the range of X and Y is called the joint probability distribution of X and Y .*

THEOREM A bivariate function can serve as the joint probability distribution of a pair of discrete random variables X and Y if and only if its values, $f(x, y)$, satisfy the conditions

1. $f(x, y) \geq 0$ for each pair of values (x, y) within its domain;
2. $\sum_x \sum_y f(x, y) = 1$, where the double summation extends over all possible pairs (x, y) within its domain.

Example: Determine the value of k for which the function given by

$$f(x, y) = kxy \quad \text{for } x = 1, 2, 3; \quad y = 1, 2, 3$$

can serve as a joint probability distribution.

Solution

Substituting the various values of x and y , we get $f(1, 1) = k, f(1, 2) = 2k, f(1, 3) = 3k, f(2, 1) = 2k, f(2, 2) = 4k, f(2, 3) = 6k, f(3, 1) = 3k, f(3, 2) = 6k$, and $f(3, 3) = 9k$. To satisfy the first condition of Theorem the constant k must be nonnegative, and to satisfy the second condition,

$$k + 2k + 3k + 2k + 4k + 6k + 3k + 6k + 9k = 1$$

so that $36k = 1$ and $k = \frac{1}{36}$.

JOINT PROBABILITY DENSITY FUNCTION:

DEFINITION **JOINT PROBABILITY DENSITY FUNCTION.** A bivariate function with values $f(x,y)$ defined over the xy -plane is called a *joint probability density function* of the continuous random variables X and Y if and only if

$$P(X, Y) \in A = \iint_A f(x,y) dx dy$$

for any region A in the xy -plane.

THEOREM A bivariate function can serve as a joint probability density function of a pair of continuous random variables X and Y if its values, $f(x,y)$, satisfy the conditions

1. $f(x,y) \geq 0$ for $-\infty < x < \infty$, $-\infty < y < \infty$;
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$.

EXAMPLE

Given the joint probability density function

$$f(x, y) = \begin{cases} \frac{3}{5}x(y+x) & \text{for } 0 < x < 1, 0 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

of two random variables X and Y , find $P[(X, Y) \in A]$, where A is the region $\{(x, y) | 0 < x < \frac{1}{2}, 1 < y < 2\}$.

Solution

$$\begin{aligned} P[(X, Y) \in A] &= P\left(0 < X < \frac{1}{2}, 1 < Y < 2\right) \\ &= \int_1^2 \int_0^{\frac{1}{2}} \frac{3}{5}x(y+x) dx dy \\ &= \int_1^2 \left. \frac{3x^2y}{10} + \frac{3x^3}{15} \right|_{x=0}^{x=\frac{1}{2}} dy \\ &= \int_1^2 \left(\frac{3y}{40} + \frac{1}{40} \right) dy = \left. \frac{3y^2}{80} + \frac{y}{40} \right|_1^2 \\ &= \frac{11}{80} \end{aligned}$$

JOINT DISTRIBUTION FUNCTION:

DEFINITION JOINT DISTRIBUTION FUNCTION. *If X and Y are discrete random variables, the function given by*

$$F(x, y) = P(X \leq x, Y \leq y) = \sum_{s \leq x} \sum_{t \leq y} f(s, t) \quad \text{for } -\infty < x < \infty \\ -\infty < y < \infty$$

where $f(s, t)$ is the value of the joint probability distribution of X and Y at (s, t) , is called the joint distribution function, or the joint cumulative distribution of X and Y .

Example: For the caplets example, find $F(1,1)$.

Solution

$$\begin{aligned} F(1, 1) &= P(X \leq 1, Y \leq 1) \\ &= f(0, 0) + f(0, 1) + f(1, 0) + f(1, 1) \\ &= \frac{1}{6} + \frac{2}{9} + \frac{1}{3} + \frac{1}{6} \\ &= \frac{8}{9} \end{aligned}$$

JOINT DISTRIBUTION FUNCTION:

DEFINITION **JOINT DISTRIBUTION FUNCTION.** *If X and Y are continuous random variables, the function given by*

$$F(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^y \int_{-\infty}^x f(s, t) ds dt \quad \text{for } -\infty < x < \infty, \\ -\infty < y < \infty$$

where $f(s, t)$ is the joint probability density of X and Y at (s, t) , is called the joint distribution function of X and Y .

EXAMPLE

If the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} x + y & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the joint distribution function of these two random variables.

Solution

If either $x < 0$ or $y < 0$, it follows immediately that $F(x, y) = 0$. For $0 < x < 1$ and $0 < y < 1$ (Region I of Figure 10), we get

$$F(x, y) = \int_0^y \int_0^x (s+t) ds dt = \frac{1}{2}xy(x+y)$$

for $x > 1$ and $0 < y < 1$ (Region II of Figure 10), we get

$$F(x, y) = \int_0^y \int_0^1 (s+t) ds dt = \frac{1}{2}y(y+1)$$

for $0 < x < 1$ and $y > 1$ (Region III of Figure 10), we get

$$F(x, y) = \int_0^1 \int_0^x (s+t) ds dt = \frac{1}{2}x(x+1)$$

and for $x > 1$ and $y > 1$ (Region IV of Figure 10), we get

$$F(x, y) = \int_0^1 \int_0^1 (s+t) ds dt = 1$$

Since the joint distribution function is everywhere continuous, the boundaries between any two of these regions can be included in either one, and we can write

$$F(x, y) = \begin{cases} 0 & \text{for } x \leq 0 \text{ or } y \leq 0 \\ \frac{1}{2}xy(x+y) & \text{for } 0 < x < 1, 0 < y < 1 \\ \frac{1}{2}y(y+1) & \text{for } x \geq 1, 0 < y < 1 \\ \frac{1}{2}x(x+1) & \text{for } 0 < x < 1, y \geq 1 \\ 1 & \text{for } x \geq 1, y \geq 1 \end{cases}$$

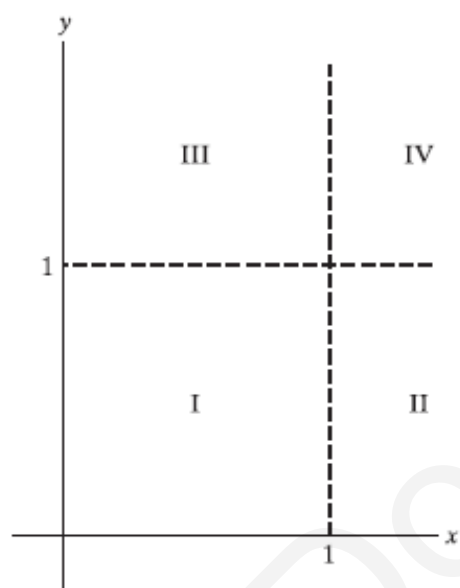
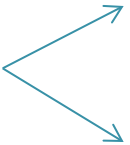
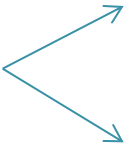


Figure 10. Diagram for Example

$f(x,y)$  Discrete $\longrightarrow$ Joint probability distribution
Continuous $\longrightarrow$ Joint probability density function

$F(x,y)$  Discrete $\longrightarrow$ Joint distribution function
(or joint cumulative distribution)
Continuous $\longrightarrow$ Joint distribution function

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y)$$

$$\frac{\partial}{\partial x} F(x, y) = \int_{-\infty}^y f(x, t) \partial t$$

$$\frac{\partial}{\partial y} F(x, y) = \int_{-\infty}^x f(s, y) \partial s$$

EXAMPLE

Find the joint probability density of the two random variables X and Y whose joint distribution function is given by

$$F(x, y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}) & \text{for } x > 0 \text{ and } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Also use the joint probability density to determine $P(1 < X < 3, 1 < Y < 2)$.

Solution

Since partial differentiation yields

$$\frac{\partial^2}{\partial x \partial y} F(x, y) = e^{-(x+y)}$$

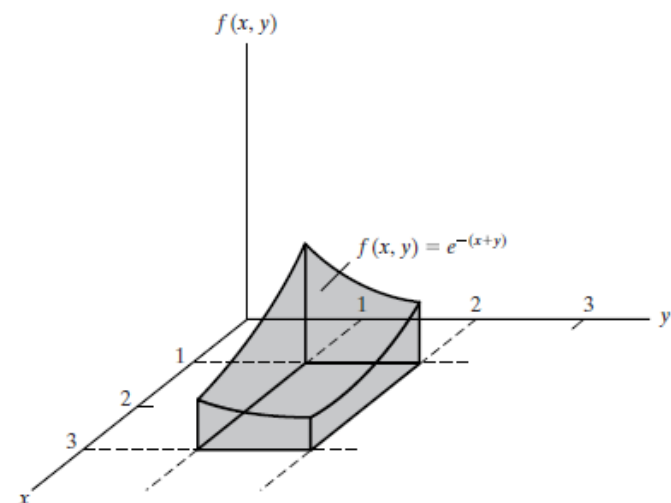
for $x > 0$ and $y > 0$ and 0 elsewhere, we find that the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} e^{-(x+y)} & \text{for } x > 0 \text{ and } y > 0 \\ 0 & \text{elsewhere} \end{cases}$$

Thus, integration yields

$$\begin{aligned} \int_1^2 \int_1^3 e^{-(x+y)} dx dy &= (e^{-1} - e^{-3})(e^{-1} - e^{-2}) \\ &= e^{-2} - e^{-3} - e^{-4} + e^{-5} \\ &= 0.074 \end{aligned}$$

for $P(1 < X < 3, 1 < Y < 2)$.



GENERALIZATION

If $X_1, X_2, \dots, X_n$ are discrete random variables, their joint probability distribution is

$$f(X_1, X_2, \dots, X_n) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n)$$

and their joint distribution function is

$$F(X_1, X_2, \dots, X_n) = P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n)$$

In the continuous case;

$$F(X_1, X_2, \dots, X_n) = \int_{-\infty}^{x_n} \dots \int_{-\infty}^{x_2} \int_{-\infty}^{x_1} f(t_1, t_2, \dots, t_n) dt_1 dt_2 \dots dt_n$$

and,

$$f(x_1, x_2, \dots, x_n) = \frac{\partial^n}{\partial x_1 \partial x_2 \dots \partial x_n} F(x_1, x_2, \dots, x_n)$$

EXAMPLE

If the joint probability distribution of three discrete random variables X , Y , and Z is given by

$$f(x, y, z) = \frac{(x+y)z}{63} \quad \text{for } x = 1, 2; \quad y = 1, 2, 3; \quad z = 1, 2$$

find $P(X = 2, Y + Z \leq 3)$.

Solution

$$\begin{aligned} P(X = 2, Y + Z \leq 3) &= f(2, 1, 1) + f(2, 1, 2) + f(2, 2, 1) \\ &= \frac{3}{63} + \frac{6}{63} + \frac{4}{63} \\ &= \frac{13}{63} \end{aligned}$$

EXAMPLE

If the trivariate probability density of X_1, X_2 , and X_3 is given by

$$f(x_1, x_2, x_3) = \begin{cases} (x_1 + x_2)e^{-x_3} & \text{for } 0 < x_1 < 1, 0 < x_2 < 1, x_3 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find $P[(X_1, X_2, X_3) \in A]$, where A is the region

$$\left\{ (x_1, x_2, x_3) \mid 0 < x_1 < \frac{1}{2}, \frac{1}{2} < x_2 < 1, x_3 < 1 \right\}$$

Solution

$$\begin{aligned} P[(X_1, X_2, X_3) \in A] &= P\left(0 < X_1 < \frac{1}{2}, \frac{1}{2} < X_2 < 1, X_3 < 1\right) \\ &= \int_0^1 \int_{\frac{1}{2}}^1 \int_0^{\frac{1}{2}} (x_1 + x_2)e^{-x_3} dx_1 dx_2 dx_3 \\ &= \int_0^1 \int_{\frac{1}{2}}^1 \left(\frac{1}{8} + \frac{x_2}{2}\right) e^{-x_3} dx_2 dx_3 \\ &= \int_0^1 \frac{1}{4} e^{-x_3} dx_3 \\ &= \frac{1}{4}(1 - e^{-1}) = 0.158 \end{aligned}$$

MARGINAL DISTRIBUTION:

EXAMPLE

In the caplet example, we derived the joint probability distribution of two random variables X and Y , the number of aspirin caplets and the number of sedative caplets included among two caplets drawn at random from a bottle containing three aspirin, two sedative, and four laxative caplets. Find the probability distribution of X alone and that of Y alone.

Solution

The results of caplet example are shown in the following table, together with the marginal totals, that is, the totals of the respective rows and columns:

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	

The column totals are the probabilities that X will take on the values 0, 1, and 2. In other words, they are the values

$$g(x) = \sum_{y=0}^2 f(x, y) \quad \text{for } x = 0, 1, 2$$

of the probability distribution of X . By the same token, the row totals are the values

$$h(y) = \sum_{x=0}^2 f(x, y) \quad \text{for } y = 0, 1, 2$$

of the probability distribution of Y .

DEFINITION MARGINAL DISTRIBUTION. *If X and Y are discrete random variables and $f(x, y)$ is the value of their joint probability distribution at (x, y) , the function given by*

$$g(x) = \sum_y f(x, y)$$

for each x within the range of X is called the marginal distribution of X . Correspondingly, the function given by

$$h(y) = \sum_x f(x, y)$$

for each y within the range of Y is called the marginal distribution of Y .

When X and Y are continuous random variables, the probability distributions are replaced by probability densities, the summations are replaced by integrals, and we obtain the following definition.

DEFINITION MARGINAL DENSITY. *If X and Y are continuous random variables and $f(x,y)$ is the value of their joint probability density at (x,y) , the function given by*

$$g(x) = \int_{-\infty}^{\infty} f(x,y) dy \quad \text{for } -\infty < x < \infty$$

is called the marginal density of X . Correspondingly, the function given by

$$h(y) = \int_{-\infty}^{\infty} f(x,y) dx \quad \text{for } -\infty < y < \infty$$

is called the marginal density of Y .

EXAMPLE

Given the joint probability density

$$f(x, y) = \begin{cases} \frac{2}{3}(x + 2y) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the marginal densities of X and Y .

Solution

Performing the necessary integrations, we get

$$g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 \frac{2}{3}(x + 2y) dy = \frac{2}{3}(x + 1)$$

for $0 < x < 1$ and $g(x) = 0$ elsewhere. Likewise,

$$h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \frac{2}{3}(x + 2y) dx = \frac{1}{3}(1 + 4y)$$

for $0 < y < 1$ and $h(y) = 0$ elsewhere.

When we are dealing with more than two random variables, we can speak not only of the marginal distributions of the individual random variables, but also of the **joint marginal distributions** of several of the random variables. If the joint probability distribution of the discrete random variables $X_1, X_2, \dots$, and X_n has the values $f(x_1, x_2, \dots, x_n)$, the marginal distribution of X_1 alone is given by

$$g(x_1) = \sum_{x_2} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n)$$

for all values within the range of X_1 , the joint marginal distribution of X_1, X_2 , and X_3 is given by

$$m(x_1, x_2, x_3) = \sum_{x_4} \cdots \sum_{x_n} f(x_1, x_2, \dots, x_n)$$

for all values within the range of X_1, X_2 , and X_3 , and other marginal distributions can be defined in the same way. For the continuous case, probability distributions are replaced by probability densities, summations are replaced by integrals, and if the joint probability density of the continuous random variables $X_1, X_2, \dots$, and X_n has the values $f(x_1, x_2, \dots, x_n)$, the marginal density of X_2 alone is given by

$$h(x_2) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_1 dx_3 \cdots dx_n$$

for $-\infty < x_2 < \infty$, the joint marginal density of X_1 and X_n is given by

$$\varphi(x_1, x_n) = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} f(x_1, x_2, \dots, x_n) dx_2 dx_3 \cdots dx_{n-1}$$

for $-\infty < x_1 < \infty$ and $-\infty < x_n < \infty$, and so forth.

EXAMPLE

Considering again the trivariate probability density of Example 1

$$f(x_1, x_2, x_3) = \begin{cases} (x_1 + x_2)e^{-x_3} & \text{for } 0 < x_1 < 1, 0 < x_2 < 1, x_3 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find the joint marginal density of X_1 and X_3 and the marginal density of X_1 alone.

Solution

Performing the necessary integration, we find that the joint marginal density of X_1 and X_3 is given by

$$m(x_1, x_3) = \int_0^1 (x_1 + x_2)e^{-x_3} dx_2 = \left(x_1 + \frac{1}{2}\right)e^{-x_3}$$

for $0 < x_1 < 1$ and $x_3 > 0$ and $m(x_1, x_3) = 0$ elsewhere. Using this result, we find that the marginal density of X_1 alone is given by

$$\begin{aligned} g(x_1) &= \int_0^\infty \int_0^1 f(x_1, x_2, x_3) dx_2 dx_3 = \int_0^\infty m(x_1, x_3) dx_3 \\ &= \int_0^\infty \left(x_1 + \frac{1}{2}\right)e^{-x_3} dx_3 = x_1 + \frac{1}{2} \end{aligned}$$

for $0 < x_1 < 1$ and $g(x_1) = 0$ elsewhere.

CONDITIONAL DISTRIBUTION:

In the conditional probability of event A , given event B , as

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

provided $P(B) \neq 0$. Suppose now that A and B are the events $X = x$ and $Y = y$ so that we can write

$$\begin{aligned} P(X = x|Y = y) &= \frac{P(X = x, Y = y)}{P(Y = y)} \\ &= \frac{f(x, y)}{h(y)} \end{aligned}$$

provided $P(Y = y) = h(y) \neq 0$, where $f(x, y)$ is the value of the joint probability distribution of X and Y at (x, y) , and $h(y)$ is the value of the marginal distribution of Y at y . Denoting the conditional probability by $f(x|y)$ to indicate that x is a variable and y is fixed, let us now make the following definition.

DEFINITION CONDITIONAL DISTRIBUTION. If $f(x, y)$ is the value of the joint probability distribution of the discrete random variables X and Y at (x, y) and $h(y)$ is the value of the marginal distribution of Y at y , the function given by

$$f(x|y) = \frac{f(x, y)}{h(y)} \quad h(y) \neq 0$$

for each x within the range of X is called the **conditional distribution of X given $Y = y$** . Correspondingly, if $g(x)$ is the value of the marginal distribution of X at x , the function given by

$$w(y|x) = \frac{f(x, y)}{g(x)} \quad g(x) \neq 0$$

for each y within the range of Y is called the **conditional distribution of Y given $X = x$** .

EXAMPLE
given $Y = 1$.

Consider the caplet example, find the conditional distribution of X

Solution

Substituting the appropriate values from the table in caplet example, we find

$$f(0|1) = \frac{\frac{2}{9}}{\frac{7}{18}} = \frac{4}{7}$$

$$f(1|1) = \frac{\frac{1}{6}}{\frac{7}{18}} = \frac{3}{7}$$

$$f(2|1) = \frac{0}{\frac{7}{18}} = 0$$

DEFINITION **CONDITIONAL DENSITY.** *If $f(x, y)$ is the value of the joint density of the continuous random variables X and Y at (x, y) and $h(y)$ is the value of the marginal distribution of Y at y , the function given by*

$$f(x|y) = \frac{f(x, y)}{h(y)} \quad h(y) \neq 0$$

for $-\infty < x < \infty$, is called the conditional density of X given $Y = y$. Correspondingly, if $g(x)$ is the value of the marginal density of X at x , the function given by

$$w(y|x) = \frac{f(x, y)}{g(x)} \quad g(x) \neq 0$$

for $-\infty < y < \infty$, is called the conditional density of Y given $X = x$.

EXAMPLE

Given the joint probability density

$$f(x, y) = \begin{cases} \frac{2}{3}(x + 2y) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the conditional density of X given $Y = y$, and

use it to evaluate $P(X \leq \frac{1}{2} | Y = \frac{1}{2})$.

Solution

Using the results obtained on the previous page, we have

$$\begin{aligned} f(x|y) &= \frac{f(x, y)}{h(y)} = \frac{\frac{2}{3}(x + 2y)}{\frac{1}{3}(1 + 4y)} \\ &= \frac{2x + 4y}{1 + 4y} \end{aligned}$$

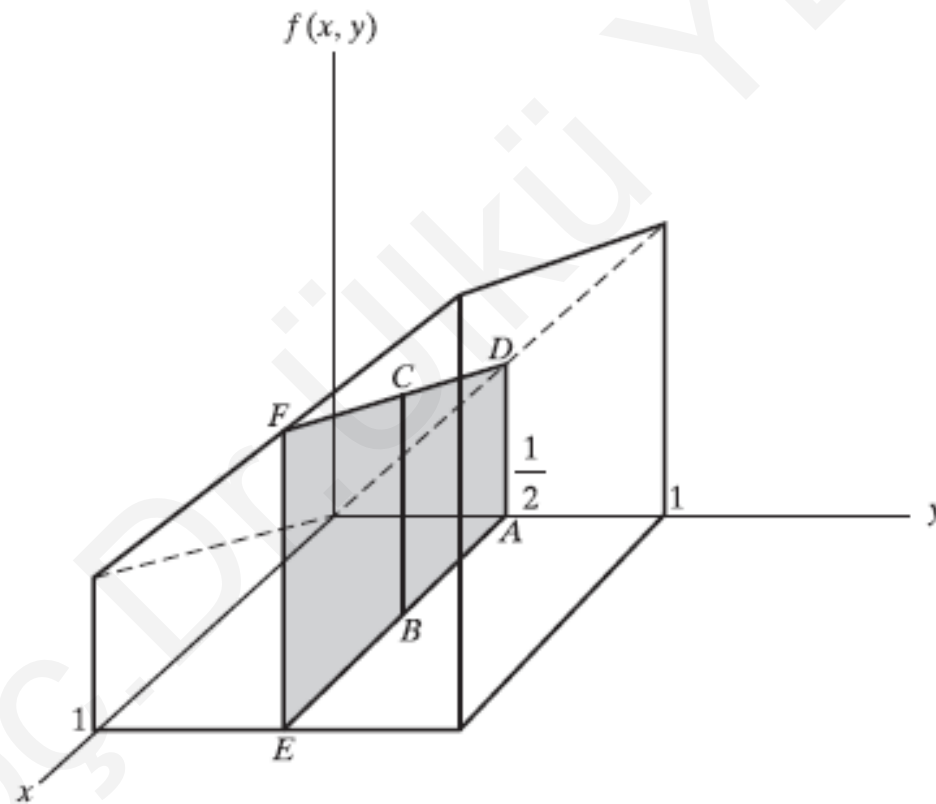
for $0 < x < 1$ and $f(x|y) = 0$ elsewhere. Now,

$$\begin{aligned} f\left(x \middle| \frac{1}{2}\right) &= \frac{2x + 4 \cdot \frac{1}{2}}{1 + 4 \cdot \frac{1}{2}} \\ &= \frac{2x + 2}{3} \end{aligned}$$

and we can write

$$P\left(X \leq \frac{1}{2} \mid Y = \frac{1}{2}\right) = \int_0^{\frac{1}{2}} \frac{2x+2}{3} dx = \frac{5}{12}$$

It is of interest to note that in Figure this probability is given by the ratio of the area of trapezoid $ABCD$ to the area of trapezoid $AEFD$.



EXAMPLE

Given the joint probability density

$$f(x, y) = \begin{cases} 4xy & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the marginal densities of X and Y and the conditional density of X given $Y = y$.

Solution

Performing the necessary integrations, we get

$$\begin{aligned} g(x) &= \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 4xy dy \\ &= 2xy^2 \Big|_{y=0}^{y=1} = 2x \end{aligned}$$

for $0 < x < 1$, and $g(x) = 0$ elsewhere; also

$$\begin{aligned} h(y) &= \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 4xy dx \\ &= 2x^2y \Big|_{x=0}^{x=1} = 2y \end{aligned}$$

for $0 < y < 1$, and $h(y) = 0$ elsewhere. Then, substituting into the formula for a conditional density, we get

$$f(x|y) = \frac{f(x, y)}{h(y)} = \frac{4xy}{2y} = 2x$$

for $0 < x < 1$, and $f(x|y) = 0$ elsewhere.

The Multivariate Case

When we are dealing with more than two random variables, whether continuous or discrete, we can consider various different kinds of conditional distributions or densities. For instance, if $f(x_1, x_2, x_3, x_4)$ is the value of the joint distribution of the discrete random variables X_1, X_2, X_3 , and X_4 at (x_1, x_2, x_3, x_4) , we can write

$$p(x_3|x_1, x_2, x_4) = \frac{f(x_1, x_2, x_3, x_4)}{g(x_1, x_2, x_4)} \quad g(x_1, x_2, x_4) \neq 0$$

for the value of the conditional distribution of X_3 at x_3 given $X_1 = x_1, X_2 = x_2$, and $X_4 = x_4$, where $g(x_1, x_2, x_4)$ is the value of the joint marginal distribution of X_1, X_2 , and X_4 at (x_1, x_2, x_4) . We can also write

$$q(x_2, x_4|x_1, x_3) = \frac{f(x_1, x_2, x_3, x_4)}{m(x_1, x_3)} \quad m(x_1, x_3) \neq 0$$

for the value of the **joint conditional distribution** of X_2 and X_4 at (x_2, x_4) given $X_1 = x_1$ and $X_3 = x_3$, or

$$r(x_2, x_3, x_4|x_1) = \frac{f(x_1, x_2, x_3, x_4)}{b(x_1)} \quad b(x_1) \neq 0$$

for the value of the joint conditional distribution of X_2, X_3 , and X_4 at (x_2, x_3, x_4) given $X_1 = x_1$.

Ex:

$$f(x_1, x_2, x_3) = \begin{cases} (x_1 + x_2)e^{-x_3}, & 0 < x_1 < 1, 0 < x_2 < 1, x_3 > 0 \\ 0, & \text{elsewhere} \end{cases}$$

find $f(x_2|x_1, x_3)$ and $P\left(x_2 \leq \frac{1}{2} \middle| x_1 = \frac{1}{2}, x_3 = 1\right)$

Solution:

$$f(x_2|x_1, x_3) = \frac{f(x_1, x_2, x_3)}{m(x_1, x_3)} = \frac{(x_1 + x_2)e^{-x_3}}{\left(x_1 + \frac{1}{2}\right)e^{-x_3}} = \frac{x_1 + x_2}{x_1 + \frac{1}{2}} \quad \text{for } 0 < x_2 < 1$$

$$f(x_2|x_1 = \frac{1}{2}, x_3 = 1) = \frac{x_1 + x_2}{x_1 + \frac{1}{2}} = \frac{\frac{1}{2} + x_2}{1}$$

$$P\left(x_2 \leq \frac{1}{2} \middle| x_1 = \frac{1}{2}, x_3 = 1\right) = \int_0^{1/2} \left(\frac{1}{2} + x_2\right) dx_2 = \frac{3}{8}$$

INDEPENDENCE

DEFINITION . INDEPENDENCE OF DISCRETE RANDOM VARIABLES. If $f(x_1, x_2, \dots, x_n)$ is the value of the joint probability distribution of the discrete random variables $X_1, X_2, \dots, X_n$ at $(x_1, x_2, \dots, x_n)$ and $f_i(x_i)$ is the value of the marginal distribution of X_i at x_i for $i = 1, 2, \dots, n$, then the n random variables are independent if and only if

$$f(x_1, x_2, \dots, x_n) = f_1(x_1) \cdot f_2(x_2) \cdot \dots \cdot f_n(x_n)$$

for all $(x_1, x_2, \dots, x_n)$ within their range.

Theorem: If random variables X and Y are independent. Conditional probability functions are equal to marginal probability functions

$$P(x|y) = P(x), \quad P(y) \neq 0$$

$$P(y|x) = P(y), \quad P(x) \neq 0$$

Proof: If random variables X and Y are independent
 $P(x,y)=P(x)P(y)$

$$P(x|y) = \frac{P(x, y)}{P(y)} = \frac{P(x)P(y)}{P(y)} = P(x)$$

EXAMPLE

Considering n independent flips of a balanced coin, let X_i be the number of heads (0 or 1) obtained in the i th flip for $i = 1, 2, \dots, n$. Find the joint probability distribution of these n random variables.

Solution

Since each of the random variables X_i , for $i = 1, 2, \dots, n$, has the probability distribution

$$f_i(x_i) = \frac{1}{2} \quad \text{for } x_i = 0, 1$$

and the n random variables are independent, their joint probability distribution is given by

$$\begin{aligned} f(x_1, x_2, \dots, x_n) &= f_1(x_1) \cdot f_2(x_2) \cdot \dots \cdot f_n(x_n) \\ &= \frac{1}{2} \cdot \frac{1}{2} \cdot \dots \cdot \frac{1}{2} = \left(\frac{1}{2}\right)^n \end{aligned}$$

where $x_i = 0$ or 1 for $i = 1, 2, \dots, n$.

EXAMPLE :

Given the independent random variables X_1 , X_2 , and X_3 with the probability densities

$$f_1(x_1) = \begin{cases} e^{-x_1} & \text{for } x_1 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$f_2(x_2) = \begin{cases} 2e^{-2x_2} & \text{for } x_2 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

$$f_3(x_3) = \begin{cases} 3e^{-3x_3} & \text{for } x_3 > 0 \\ 0 & \text{elsewhere} \end{cases}$$

find their joint probability density, and use it to evaluate the probability $P(X_1 + X_2 \leq 1, X_3 > 1)$.

Solution

$$\begin{aligned} f(x_1, x_2, x_3) &= f_1(x_1) \cdot f_2(x_2) \cdot f_3(x_3) \\ &= e^{-x_1} \cdot 2e^{-2x_2} \cdot 3e^{-3x_3} \\ &= 6e^{-x_1-2x_2-3x_3} \end{aligned}$$

for $x_1 > 0, x_2 > 0, x_3 > 0$, and $f(x_1, x_2, x_3) = 0$ elsewhere. Thus,

$$\begin{aligned} P(X_1 + X_2 \leq 1, X_3 > 1) &= \int_1^\infty \int_0^1 \int_0^{1-x_2} 6e^{-x_1-2x_2-3x_3} dx_1 dx_2 dx_3 \\ &= (1 - 2e^{-1} + e^{-2})e^{-3} \\ &= 0.020 \end{aligned}$$

THE EXPECTED VALUE FOR MULTIVARIATE CASE

THEOREM If X and Y are discrete random variables and $f(x, y)$ is the value of their joint probability distribution at (x, y) , the expected value of $g(X, Y)$ is

$$E[g(X, Y)] = \sum_x \sum_y g(x, y) \cdot f(x, y)$$

Correspondingly, if X and Y are continuous random variables and $f(x, y)$ is the value of their joint probability density at (x, y) , the expected value of $g(X, Y)$ is

$$E[g(X, Y)] = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f(x, y) dx dy$$

EXAMPLE

Consider the caplets example. Find the expected value of $g(X, Y) = X + Y$.

Solution

$$\begin{aligned} E(X + Y) &= \sum_{x=0}^2 \sum_{y=0}^2 (x + y) \cdot f(x, y) \\ &= (0 + 0) \cdot \frac{1}{6} + (0 + 1) \cdot \frac{2}{9} + (0 + 2) \cdot \frac{1}{36} + (1 + 0) \cdot \frac{1}{3} \\ &\quad + (1 + 1) \cdot \frac{1}{6} + (2 + 0) \cdot \frac{1}{12} \\ &= \frac{10}{9} \end{aligned}$$

$$E(XY) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy \cdot f(x, y) dx dy$$

$$E(X) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xf(x, y) dx dy = \int_{-\infty}^{+\infty} xf(x) dx = \mu_x$$

$$E(Y) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} yf(x, y) dx dy = \int_{-\infty}^{+\infty} yf(y) dy = \mu_y$$

EXAMPLE

If the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} \frac{2}{7}(x+2y) & \text{for } 0 < x < 1, 1 < y < 2 \\ 0 & \text{elsewhere} \end{cases}$$

find the expected value of $g(X, Y) = X/Y^3$.

Solution

$$\begin{aligned} E(X/Y^3) &= \int_1^2 \int_0^1 \frac{2x(x+2y)}{7y^3} dx dy \\ &= \frac{2}{7} \int_1^2 \left(\frac{1}{3y^3} + \frac{1}{y^2} \right) dy \\ &= \frac{15}{84} \end{aligned}$$

THEOREM If $c_1, c_2, \dots$, and c_n are constants, then

$$E \left[\sum_{i=1}^n c_i g_i(X_1, X_2, \dots, X_k) \right] = \sum_{i=1}^n c_i E[g_i(X_1, X_2, \dots, X_k)]$$

PRODUCT MOMENTS

DEFINITION PRODUCT MOMENTS ABOUT THE ORIGIN. *The r th and s th product moment about the origin of the random variables X and Y , denoted by $\mu'_{r,s}$, is the expected value of $X^r Y^s$; symbolically,*

$$\mu'_{r,s} = E(X^r Y^s) = \sum_x \sum_y x^r y^s \cdot f(x, y)$$

for $r = 0, 1, 2, \dots$ and $s = 0, 1, 2, \dots$ when X and Y are discrete, and

$$\mu'_{r,s} = E(X^r Y^s) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^r y^s \cdot f(x, y) dx dy$$

when X and Y are continuous.

DEFINITION **PRODUCT MOMENTS ABOUT THE MEAN.** *The r th and s th product moment about the means of the random variables X and Y , denoted by $\mu_{r,s}$, is the expected value of $(X - \mu_X)^r (Y - \mu_Y)^s$; symbolically,*

$$\begin{aligned}\mu_{r,s} &= E[(X - \mu_X)^r (Y - \mu_Y)^s] \\ &= \sum_x \sum_y (x - \mu_X)^r (y - \mu_Y)^s \cdot f(x, y)\end{aligned}$$

for $r = 0, 1, 2, \dots$ and $s = 0, 1, 2, \dots$ when X and Y are discrete, and

$$\begin{aligned}\mu_{r,s} &= E[(X - \mu_X)^r (Y - \mu_Y)^s] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (x - \mu_X)^r (y - \mu_Y)^s \cdot f(x, y) dx dy\end{aligned}$$

when X and Y are continuous.

COVARIANCE

DEFINITION COVARIANCE. $\mu_{1,1}$ is called the covariance of X and Y , and it is denoted by σ_{XY} , $\text{cov}(X, Y)$, or $C(X, Y)$.

THEOREM

$$\sigma_{XY} = \mu'_{1,1} - \mu_X \mu_Y$$

Proof Using the various theorems about expected values, we can write

$$\begin{aligned}\sigma_{XY} &= E[(X - \mu_X)(Y - \mu_Y)] \\ &= E(XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y) \\ &= E(XY) - \mu_Y E(X) - \mu_X E(Y) + \mu_X\mu_Y \\ &= E(XY) - \mu_Y\mu_X - \mu_X\mu_Y + \mu_X\mu_Y \\ &= \mu'_{1,1} - \mu_X\mu_Y\end{aligned}$$

$$\text{Cov}(X, Y) = E(X - E(X))(Y - E(Y)) = E(XY) - E(X)E(Y)$$

$$= E(XY) - \mu_x \mu_y$$

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

$$\text{Cov}(X, X) = \text{Var}(X) = \sigma_x^2$$

EXAMPLE

Consider the caplet example.

The joint and marginal probabilities of X and Y , the numbers of aspirin and sedative caplets among two caplets drawn at random from a bottle containing three aspirin, two sedative, and four laxative caplets, are recorded as follows:

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	

Find the covariance of X and Y .

Solution

Referring to the joint probabilities given here, we get

$$\begin{aligned}\mu'_{1,1} &= E(XY) \\ &= 0 \cdot 0 \cdot \frac{1}{6} + 0 \cdot 1 \cdot \frac{2}{9} + 0 \cdot 2 \cdot \frac{1}{36} + 1 \cdot 0 \cdot \frac{1}{3} + 1 \cdot 1 \cdot \frac{1}{6} + 2 \cdot 0 \cdot \frac{1}{12} \\ &= \frac{1}{6}\end{aligned}$$

and using the marginal probabilities, we get

$$\mu_X = E(X) = 0 \cdot \frac{5}{12} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{12} = \frac{2}{3}$$

and

$$\mu_Y = E(Y) = 0 \cdot \frac{7}{12} + 1 \cdot \frac{7}{18} + 2 \cdot \frac{1}{36} = \frac{4}{9}$$

It follows that

$$\sigma_{XY} = \frac{1}{6} - \frac{2}{3} \cdot \frac{4}{9} = -\frac{7}{54}$$

The negative result suggests that the more aspirin tablets we get the fewer sedative tablets we will get, and vice versa, and this, of course, makes sense.

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	

EXAMPLE

Find the covariance of the random variables whose joint probability density is given by

$$f(x, y) = \begin{cases} 2 & \text{for } x > 0, y > 0, x + y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Solution

Evaluating the necessary integrals, we get

$$E(X) = \mu_X = \int_0^1 \int_0^{1-x} 2x \, dy \, dx = \frac{1}{3}$$

$$E(Y) = \mu_Y = \int_0^1 \int_0^{1-x} 2y \, dy \, dx = \frac{1}{3}$$

and

$$E(XY) = \int_0^1 \int_0^{1-x} 2xy \, dy \, dx = \frac{1}{12}$$

It follows that

$$\sigma_{XY} = \frac{1}{12} - \frac{1}{3} \cdot \frac{1}{3} = -\frac{1}{36}$$

THEOREM If X and Y are independent, then $E(XY) = E(X) \cdot E(Y)$ and $\sigma_{XY} = 0$.

Proof For the discrete case we have, by definition,

$$E(XY) = \sum_x \sum_y xy \cdot f(x, y)$$

Since X and Y are independent, we can write $f(x, y) = g(x) \cdot h(y)$, where $g(x)$ and $h(y)$ are the values of the marginal distributions of X and Y , and we get

$$\begin{aligned} E(XY) &= \sum_x \sum_y xy \cdot g(x)h(y) \\ &= \left[\sum_x x \cdot g(x) \right] \left[\sum_y y \cdot h(y) \right] \\ &= E(X) \cdot E(Y) \end{aligned}$$

Hence,

$$\begin{aligned} \sigma_{XY} &= \mu'_{1,1} - \mu_X \mu_Y \\ &= E(X) \cdot E(Y) - E(X) \cdot E(Y) \\ &= 0 \end{aligned}$$

Note that the independency of two variables implies a zero covariance, but a zero covariance does not necessarily imply their independence.

EXAMPLE

If the joint probability distribution of X and Y is given by

show that their covariance is zero even though the two random variables are not independent.

Solution

Using the probabilities shown in the margins, we get

$$\mu_X = (-1) \cdot \frac{1}{3} + 0 \cdot \frac{1}{3} + 1 \cdot \frac{1}{3} = 0$$

$$\mu_Y = (-1) \cdot \frac{2}{3} + 0 \cdot 0 + 1 \cdot \frac{1}{3} = -\frac{1}{3}$$

and

$$\begin{aligned} E(XY) = \mu'_{1,1} &= (-1)(-1) \cdot \frac{1}{6} + 0(-1) \cdot \frac{1}{3} + 1(-1) \cdot \frac{1}{6} + (-1)1 \cdot \frac{1}{6} + 1 \cdot 1 \cdot \frac{1}{6} \\ &= 0 \end{aligned}$$

Thus, $\sigma_{XY} = 0 - 0(-\frac{1}{3}) = 0$, the covariance is zero, but the two random variables are not independent. For instance, $f(x, y) \neq g(x) \cdot h(y)$ for $x = -1$ and $y = -1$.

	x			
	-1	0	1	
-1	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{2}{3}$
0	0	0	0	0
1	$\frac{1}{6}$	0	$\frac{1}{6}$	$\frac{1}{3}$
	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	

THEOREM If $X_1, X_2, \dots, X_n$ are independent, then

$$E(X_1 X_2 \cdots X_n) = E(X_1) \cdot E(X_2) \cdots E(X_n)$$

Definition: Variance of the sum of the two random variables X and Y

$$\text{Var}(X+Y)=\text{Var}(X)+\text{Var}(Y)+2\text{Cov}(X,Y)$$

Variance of the difference of two random variables,

$$\text{Var}(X-Y)=\text{Var}(X)+\text{Var}(Y)-2\text{Cov}(X,Y)$$

CORRELATION

Definition: The correlation of X and Y is defined as;

$$\text{corr}(X, Y) = \rho_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_x \sigma_y} = \frac{E[(X - \mu_x)(Y - \mu_y)]}{\sqrt{E[(X - \mu_x)^2]} \sqrt{E[(Y - \mu_y)^2]}}$$

$\text{corr}(X,Y)$ is dimensionless, while $\text{cov}(X,Y)$ has units.

Correlation coefficient; $-1 \leq \rho_{xy} \leq 1$

$\rho_{xy} = 0$ There is no linear correlation

$\rho_{xy} < 0$ inverse linear correlation

$\rho_{xy} > 0$ linear correlation

$\rho_{xy} = -1$ There is a perfect inverse linear correlation

$\rho_{xy} = +1$ There is a perfect linear correlation

The degree of the relationship gets stronger as it move away from zero.

Theorem: If X and Y random variables provide $Y = aX + b$ equation

$$\rho_{xy} = \begin{cases} -1, & a < 0 \\ 0, & a = 0 \\ 1, & a > 0 \end{cases}$$

$$\rho_{xy} = \rho_{yx}$$

MOMENTS OF LINEAR COMBINATIONS OF RANDOM VARIABLES

THEOREM If $X_1, X_2, \dots, X_n$ are random variables and

$$Y = \sum_{i=1}^n a_i X_i$$

where $a_1, a_2, \dots, a_n$ are constants, then

$$E(Y) = \sum_{i=1}^n a_i E(X_i)$$

and

$$\text{var}(Y) = \sum_{i=1}^n a_i^2 \cdot \text{var}(X_i) + 2 \sum_{i < j} a_i a_j \cdot \text{cov}(X_i, X_j)$$

where the double summation extends over all values of i and j , from 1 to n , for which $i < j$.

Proof

$$E(Y) = E\left(\sum_{i=1}^n a_i X_i\right) = \sum_{i=1}^n a_i E(X_i)$$

$$\begin{aligned} \text{var}(Y) &= E\left([Y - E(Y)]^2\right) = E\left\{\left[\sum_{i=1}^n a_i X_i - \sum_{i=1}^n a_i E(X_i)\right]^2\right\} \\ &= E\left\{\left[\sum_{i=1}^n a_i (X_i - \mu_i)\right]^2\right\} \end{aligned}$$

Then, expanding by means of the multinomial theorem, according to which $(a + b + c + d)^2$, for example, equals $a^2 + b^2 + c^2 + d^2 + 2ab + 2ac + 2ad + 2bc + 2bd + 2cd$, and again referring to Theorem , we get

$$\begin{aligned}\text{var}(Y) &= \sum_{i=1}^n a_i^2 E[(X_i - \mu_i)^2] + 2 \sum_{i < j} a_i a_j E[(X_i - \mu_i)(X_j - \mu_j)] \\ &= \sum_{i=1}^n a_i^2 \cdot \text{var}(X_i) + 2 \sum_{i < j} a_i a_j \cdot \text{cov}(X_i, X_j)\end{aligned}$$

Note that we have tacitly made use of the fact that $\text{cov}(X_i, X_j) = \text{cov}(X_j, X_i)$.

Since $\text{cov}(X_i, X_j) = 0$ when X_i and X_j are independent, we obtain the following corollary.

COROLLARY If the random variables $X_1, X_2, \dots, X_n$ are independent and

$Y = \sum_{i=1}^n a_i X_i$, then

$$\text{var}(Y) = \sum_{i=1}^n a_i^2 \cdot \text{var}(X_i)$$

EXAMPLE

If the random variables X , Y , and Z have the means $\mu_X = 2$, $\mu_Y = -3$, and $\mu_Z = 4$, the variances $\sigma_X^2 = 1$, $\sigma_Y^2 = 5$, and $\sigma_Z^2 = 2$, and the covariances $\text{cov}(X, Y) = -2$, $\text{cov}(X, Z) = -1$, and $\text{cov}(Y, Z) = 1$, find the mean and the variance of $W = 3X - Y + 2Z$.

Solution

$$\begin{aligned} E(W) &= E(3X - Y + 2Z) \\ &= 3E(X) - E(Y) + 2E(Z) \\ &= 3 \cdot 2 - (-3) + 2 \cdot 4 \\ &= 17 \end{aligned}$$

and

$$\begin{aligned} \text{var}(W) &= 9 \text{var}(X) + \text{var}(Y) + 4 \text{var}(Z) - 6 \text{cov}(X, Y) \\ &\quad + 12 \text{cov}(X, Z) - 4 \text{cov}(Y, Z) \\ &= 9 \cdot 1 + 5 + 4 \cdot 2 - 6(-2) + 12(-1) - 4 \cdot 1 \\ &= 18 \end{aligned}$$

THEOREM If $X_1, X_2, \dots, X_n$ are random variables and

$$Y_1 = \sum_{i=1}^n a_i X_i \quad \text{and} \quad Y_2 = \sum_{i=1}^n b_i X_i$$

where $a_1, a_2, \dots, a_n, b_1, b_2, \dots, b_n$ are constants, then

$$\text{cov}(Y_1, Y_2) = \sum_{i=1}^n a_i b_i \cdot \text{var}(X_i) + \sum_{i < j} \sum (a_i b_j + a_j b_i) \cdot \text{cov}(X_i, X_j)$$

COROLLARY If the random variables $X_1, X_2, \dots, X_n$ are **independent**, $Y_1 = \sum_{i=1}^n a_i X_i$ and $Y_2 = \sum_{i=1}^n b_i X_i$, then

$$\text{cov}(Y_1, Y_2) = \sum_{i=1}^n a_i b_i \cdot \text{var}(X_i)$$

EXAMPLE

If the random variables X , Y , and Z have the means $\mu_X = 3$, $\mu_Y = 5$, and $\mu_Z = 2$, the variances $\sigma_X^2 = 8$, $\sigma_Y^2 = 12$, and $\sigma_Z^2 = 18$, and $\text{cov}(X, Y) = 1$, $\text{cov}(X, Z) = -3$, and $\text{cov}(Y, Z) = 2$, find the covariance of

$$U = X + 4Y + 2Z \quad \text{and} \quad V = 3X - Y - Z$$

Solution

$$\text{cov}(Y_1, Y_2) = \sum_{i=1}^n a_i b_i \cdot \text{var}(X_i) + \sum_{i < j} (a_i b_j + a_j b_i) \cdot \text{cov}(X_i, X_j)$$

$$\begin{aligned} \text{cov}(U, V) &= \text{cov}(X + 4Y + 2Z, 3X - Y - Z) \\ &= 3 \text{var}(X) - 4 \text{var}(Y) - 2 \text{var}(Z) + 11 \text{cov}(X, Y) \\ &\quad + 5 \text{cov}(X, Z) - 6 \text{cov}(Y, Z) \\ &= 3 \cdot 8 - 4 \cdot 12 - 2 \cdot 18 + 11 \cdot 1 + 5(-3) - 6 \cdot 2 \\ &= -76 \end{aligned}$$

CONDITIONAL EXPECTATIONS

DEFINITION . CONDITIONAL EXPECTATION. If X is a discrete random variable, and $f(x|y)$ is the value of the conditional probability distribution of X given $Y = y$ at x , the conditional expectation of $u(X)$ given $Y = y$ is

$$E[u(X)|y] = \sum_x u(x) \cdot f(x|y)$$

Correspondingly, if X is a continuous variable and $f(x|y)$ is the value of the conditional probability distribution of X given $Y = y$ at x , the conditional expectation of $u(X)$ given $Y = y$ is

$$E[(u(X)|y)] = \int_{-\infty}^{\infty} u(x) \cdot f(x|y) dx$$

If we let $u(X) = X$ in Definition we obtain the **conditional mean** of the random variable X given $Y = y$, which we denote by

$$\mu_{X|y} = E(X|y)$$

Correspondingly, the **conditional variance** of X given $Y = y$ is

$$\begin{aligned}\sigma_{X|y}^2 &= E[(X - \mu_{X|y})^2|y] \\ &= E(X^2|y) - \mu_{X|y}^2\end{aligned}$$

where $E(X^2|y)$ is given by Definition with $u(X) = X^2$.

EXAMPLE

If the joint probability density of X and Y is given by

$$f(x, y) = \begin{cases} \frac{2}{3}(x+2y) & \text{for } 0 < x < 1, 0 < y < 1 \\ 0 & \text{elsewhere} \end{cases}$$

find the conditional mean and the conditional variance of X given $Y = \frac{1}{2}$.

Solution

For these random variables the conditional density of X given $Y = y$ is

$$f(x|y) = \begin{cases} \frac{2x+4y}{1+4y} & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

so that

$$f\left(x\left|\frac{1}{2}\right.\right) = \begin{cases} \frac{2}{3}(x+1) & \text{for } 0 < x < 1 \\ 0 & \text{elsewhere} \end{cases}$$

Thus, $\mu_{X|\frac{1}{2}}$ is given by

$$\begin{aligned} E\left(X\left|\frac{1}{2}\right.\right) &= \int_0^1 \frac{2}{3}x(x+1) dx \\ &= \frac{5}{9} \end{aligned}$$

Next we find

$$\begin{aligned} E\left(X^2\left|\frac{1}{2}\right.\right) &= \int_0^1 \frac{2}{3}x^2(x+1) dx \\ &= \frac{7}{18} \end{aligned}$$

and it follows that

$$\sigma_{X|\frac{1}{2}}^2 = \frac{7}{18} - \left(\frac{5}{9}\right)^2 = \frac{13}{162}$$

Example: Consider the caplets example.

- Find the conditional mean of X given $Y=1$.
- Find the conditional mean of Y given $X=1$.
- If $g(X,Y)=X+Y$, find $E(X+Y)=?$

		x			
		0	1	2	
y	0	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{12}$	$\frac{7}{12}$
	1	$\frac{2}{9}$	$\frac{1}{6}$		$\frac{7}{18}$
	2	$\frac{1}{36}$			$\frac{1}{36}$
		$\frac{5}{12}$	$\frac{1}{2}$	$\frac{1}{12}$	